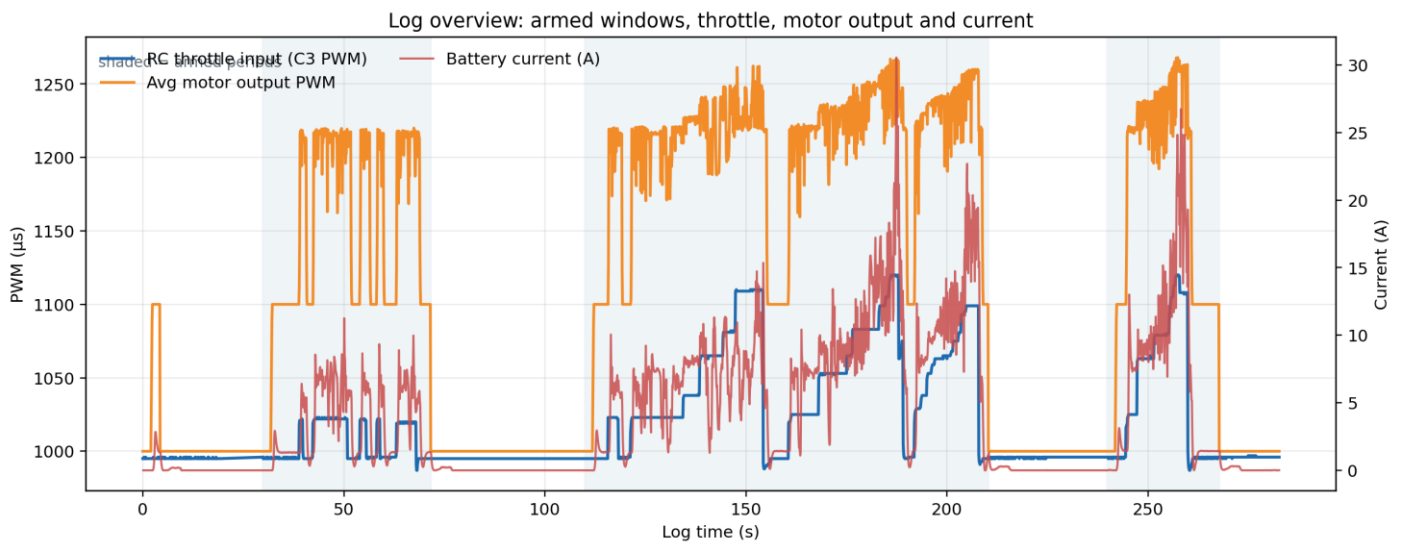
What is inside the log?



What I checked

DataFlash messages: ATT, RATE, PIDR/P/Y, VIBE, IMU, RCOU/RCIN, BAT, GPS, EKF and MSG events.

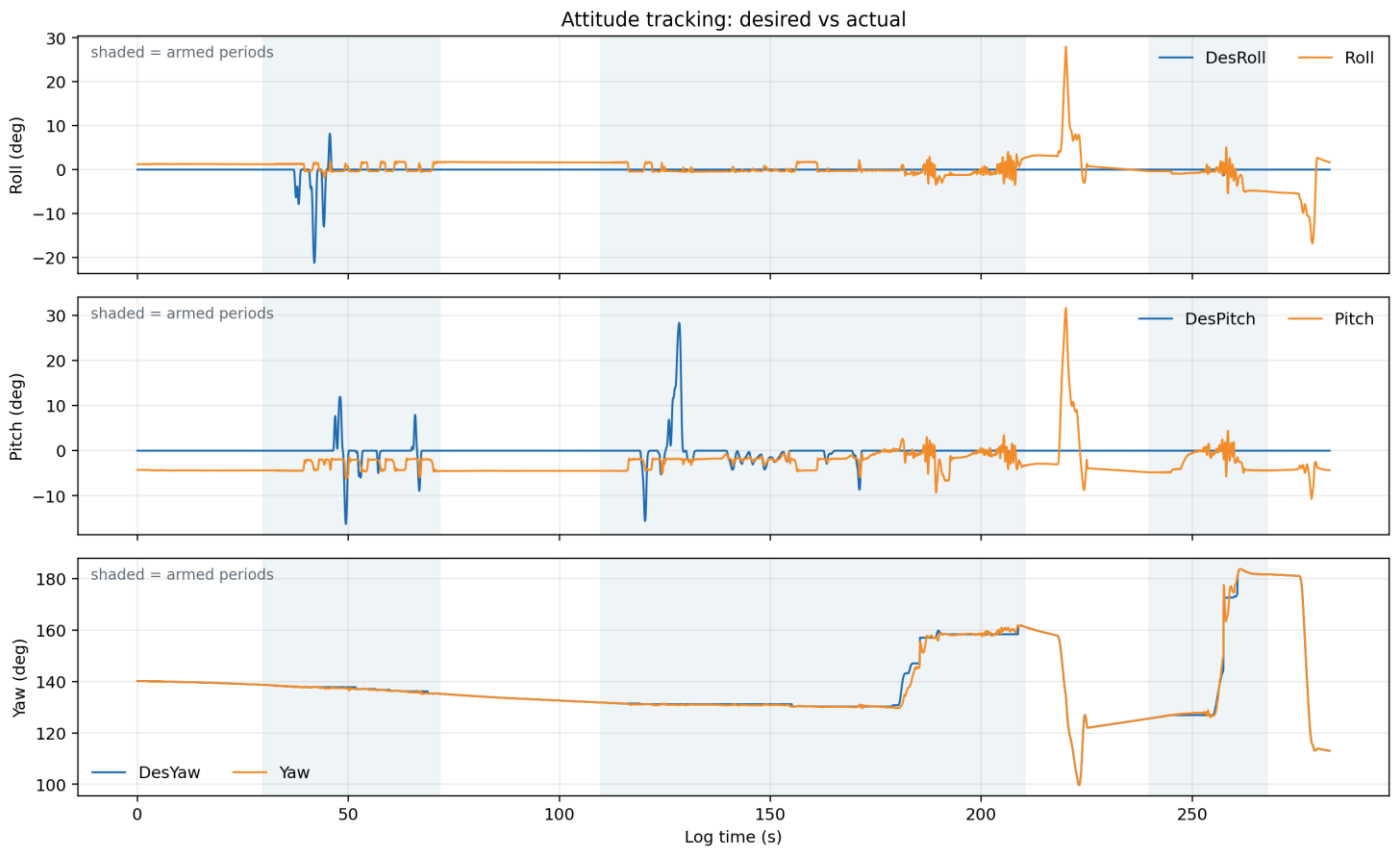
Important limitation

This is not a clean hover/flight-tuning log. The vehicle is mostly low-throttle / ground-run in Stabilize mode, so PID conclusions are first-look only.

Flight evidence

3 armed windows: 41.8 s, 100.5 s, 27.9 s. RC throttle stayed low; GPS speed was almost stationary.

Attitude response: desired vs actual



- Roll and pitch actual angles do not sit exactly at desired zero during neutral-stick periods.

- **Pitch error is larger than roll** in the overall log; this can come from ground tilt, level calibration, CG bias, or motor/prop asymmetry.

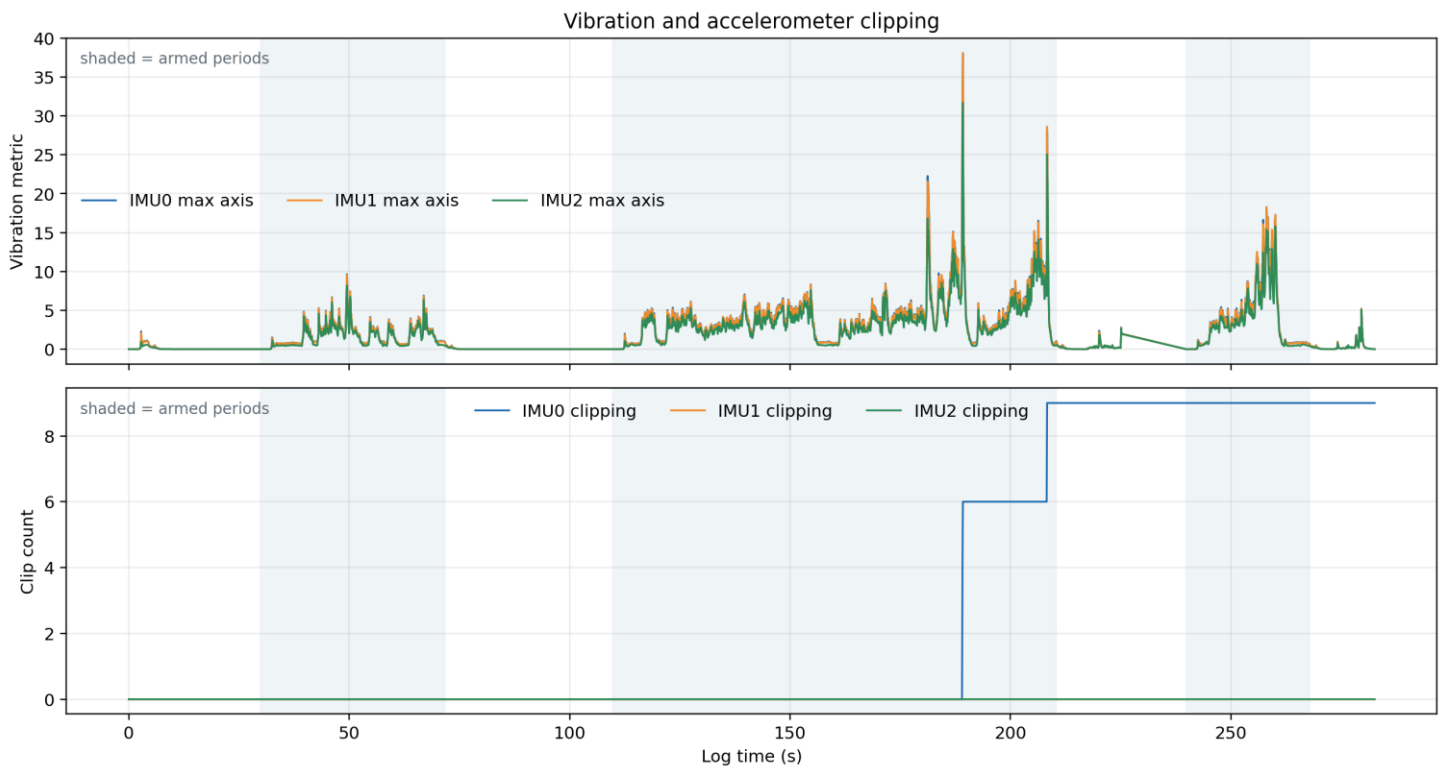
- Yaw mostly follows desired yaw, but yaw-related EKF events make compass review mandatory before autonomous modes.

- Because this **was not a stable hover**, we should not judge final attitude gains yet.

Measured from armed periods

Roll attitude MAE ≈ 1.14°
Pitch attitude MAE ≈ 2.67°
Yaw error MAE ≈ 0.55°

Vibration and filter status



- Vibration peaks are significant during armed motor-run sections; peak values reached ~34–38 on IMU0/IMU1 axes.

- IMU0 clipping reached 9 counts. Even if not continuous, any clipping is a warning sign before aggressive tuning.

- INS_GYRO_FILTER = 20 Hz, INS_ACCEL_FILTER = 10 Hz.

- ATC_RAT_RLL/PIT_FLTD and FLTT are at 10 Hz, lower than default 20 Hz; this can protect from noise but may slow response.

- **INS_HNTCH_ENABLE = 0** and **INS_LOG_BAT_MASK = 0**; no harmonic-notch validation from this log.

Suspected work item

Enable raw IMU batch logging and configure harmonic notch after measuring motor/prop vibration frequencies.

Recommended tuning plan

1 Mechanical baseline

Prop balance, motor health, frame stiffness, FC mount, CG, motor order.

2 Sensor baseline

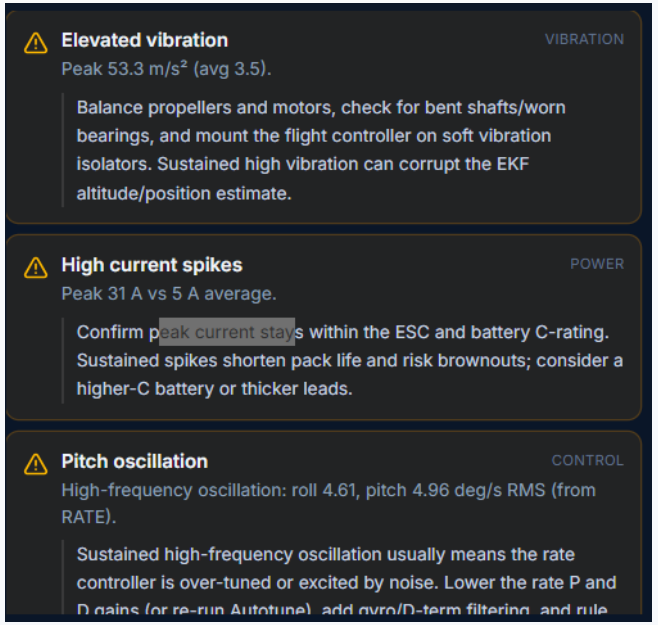
Accel level, compass calibration/reboot, EKF yaw stability, battery/current calibration.

3 Filter baseline

Raw IMU batch log, FFT review, harmonic notch, then validate noise margins.

4 PID tuning

Controlled hover, step inputs, then AutoTune/manual gain refinement.



- Ask the client for a proper hover/Loiter log with moderate stick inputs, no wind if possible, and LOG_BITMASK/batch logging enabled for IMU filter analysis.
- Do not increase P/D gains until compass and vibration issues are resolved.
- After filters are validated, tune rate loops first, then attitude/position controllers.
- For agricultural/heavy UAVs, repeat tuning at representative payload and battery weight.